Exercise 4: Worked solutions

Electrical Engineering I · Module 92315 · Exercise 4

From Fundamentals to AI Apocalypse · personal study guide

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Made by Thanate Amornjiraporn (Focus)

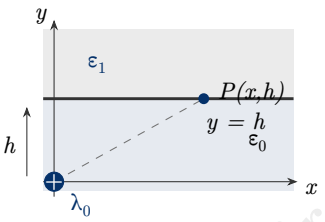
Try each part before opening the answer. We apply the interface conditions to the measured field, then use cylindrical Gauss surfaces and voltage integration for the layered coaxial geometry.

Task 9: Field at a flat dielectric boundary

An infinite line charge of density λ_0 lies along the z-axis. The lower half-space y < h is air with permittivity ϵ_0, and the upper half-space y ≥ h has permittivity ϵ_1. In the air, measurements give the field below.

$$E_0(\mathrm{x},\mathrm{y}) = \frac{\lambda_0}{\pi(\varepsilon_0 + \varepsilon_1)(\mathrm{x}^2 + \mathrm{y}^2)}\left[\mathrm{x}e_\mathrm{x} + \frac{\varepsilon_1}{\varepsilon_0}\mathrm{y}e_\mathrm{y}\right]$$

Connected topic: Material boundary conditions (§1.11) keep tangential E continuous and normal D continuous when no free surface charge is present.



Resolve the measured field into interface-parallel and interface-normal parts. The x-component stays continuous; the y-component changes with permittivity.

9: Electric field in the dielectric at y = h

Try first: evaluate the measured air field at the interface, then apply the tangential-E and normal-D boundary conditions component by component.

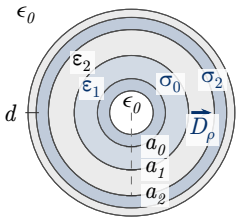
Answers: see "Task 9: Field at a flat dielectric boundary" under Answers at the end.

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Task 10: Two-dielectric coaxial capacitor

Two infinitely long concentric metal cylinders share the z-axis. The inner metal's outer radius is a_0; the outer metal's inner radius is $a_2 > a_0$. Between them, permittivity is ϵ_1 for $a_0 < \rho < a_1$ and ϵ_2 for $a_1 < \rho < a_2$. The conductor walls each have thickness d; outside the conductors is material ϵ_0. The only charges are σ_0 at the inner conductor's outer surface and σ_2 at the outer conductor's inner surface.

Connected topics: Gauss's law (§1.10) finds the radial flux; capacitance (§1.13) follows from charge per length divided by voltage.



Track each cylindrical boundary by radius. The field runs radially through ε_1 then ε_2; free surface charges sit on the two metal faces bordering the dielectric gap.

In the dielectric gap, symmetry gives $D = D_\rho(\rho)e_\rho$. The sheet fixes the voltage direction: $U_{20} = \int$ from P_2 (outer) to P_0 (inner) of $E \cdot ds$, matching the figure's arrow from the outer to the inner terminal and the course convention $U_{20} = \varphi_2 - \varphi_0$. The path therefore runs inward.

10a: Flux density, field, and outer surface charge

Try first: apply cylindrical Gauss surfaces in each dielectric, then use continuity of normal D at $\rho = a_1$ and the conductor boundary at $\rho = a_2$.

Answers: see "Task 10: Two-dielectric coaxial capacitor" under Answers at the end.

10b: Field and σ_0 as functions of U_{20}

Try first: integrate E along the path the sheet gives, inward from P_2 at a_2 to P_0 at a_0, splitting at a_1, then solve the voltage relation for σ_0.

Answers: see "Task 10: Two-dielectric coaxial capacitor" under Answers at the end.

10c: Capacitance for length Δz

Try first: use the inner conductor's charge over length Δz and divide by the voltage from 10b.

Answers: see "Task 10: Two-dielectric coaxial capacitor" under Answers at the end.

10d: Field inside the inner conductor and outside the outer one

Try first: use Gauss's law and the conductor condition $D = 0$, remembering that the two facing surface charges cancel per unit length.

Answers: see "Task 10: Two-dielectric coaxial capacitor" under Answers at the end.

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Answers

Task 9: Field at a flat dielectric boundary

9 · Apply both interface conditions

1. The interface is horizontal, so its normal is along y and its tangent is along x. There is no free charge sheet in the problem. Therefore:

$$E_{1x} = E_{0x}; \quad D_{1y} = D_{0y}, \quad \text{equivalently } \varepsilon_1 E_{1y} = \varepsilon_0 E_{0y}$$

2. First evaluate the given air field at y = h. Its tangential component is:

$$E_{0x}(x, h) = \frac{\lambda_0 x}{\pi(\varepsilon_0 + \varepsilon_1)(x^2 + h^2)}$$

3. Keep that x-component unchanged across the boundary:

$$E_{1x}(x, h) = E_{0x}(x, h)$$

4. The measured normal component contains the factor $\varepsilon_1/\varepsilon_0$. Multiplying it by $\varepsilon_0/\varepsilon_1$ to preserve D cancels that factor:

$$E_{1y}(x, h) = \frac{\varepsilon_0}{\varepsilon_1} E_{0y}(x, h) = \frac{\lambda_0 h}{\pi(\varepsilon_0 + \varepsilon_1)(x^2 + h^2)}$$

5. **Result:** combine the two components into the dielectric-side vector:

$$E_1(x, h) = \frac{\lambda_0}{\pi(\varepsilon_0 + \varepsilon_1)(x^2 + h^2)} [x e_x + h e_y]$$

Check: the tangential component is unchanged; the normal flux density matches on both sides. With SI inputs, E is in V/m (equivalently N/C).

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Task 10: Two-dielectric coaxial capacitor

10a · Use enclosed charge per unit length

1. Choose a coaxial Gaussian cylinder of radius ρ and length Δz, with $a_0 < \rho < a_2$. Radial D is constant over its curved side; its end caps contribute no flux. The enclosed free charge is the inner surface charge on length Δz:

$$D_\rho(\rho)\, 2\pi\rho\, \Delta z \;=\; \sigma_0\, 2\pi a_0\, \Delta z$$

2. Cancel the common factors. No free charge lies at the dielectric boundary $\rho = a_1$, so normal D has the same value in both layers:

$$D(\rho) \;=\; \frac{\sigma_0 a_0}{\rho}\, e_\rho, \qquad a_0 \;<\; \rho \;<\; a_2$$

3. Use $D = \varepsilon E$ separately in each material:

$$E(\rho) \;=\; \frac{\sigma_0 a_0}{\varepsilon_1 \rho} e_\rho, \qquad a_0 \;<\; \rho \;<\; a_1$$

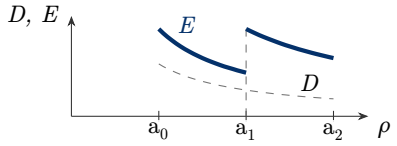
$$E(\rho) \;=\; \frac{\sigma_0 a_0}{\varepsilon_2 \rho} e_\rho, \qquad a_1 \;<\; \rho \;<\; a_2$$

4. At the inner face of the outer metal cylinder, the normal from the conductor into the gap points inward, $-e_\rho$. Thus $\sigma_2 = D \cdot (-e_\rho)$ at $\rho = a_2$:

$$\sigma_2 \;=\; -\frac{\sigma_0 a_0}{a_2}$$

Result: D is radial and continuous through the dielectric interface; E changes magnitude inversely with permittivity. The two metal-face charges have opposite signs.

Check: $\sigma_2\, 2\pi a_2 \Delta z = -\sigma_0\, 2\pi a_0 \Delta z$, so the two facing surfaces carry equal and opposite charge.



D runs on, E jumps. Drawn for $\varepsilon_2 = \varepsilon_1/2$; the two curves are not to the same scale. D (dashed) falls smoothly as $1/\rho$ across the interface at a_1, while $E = D/\varepsilon$ jumps there by the ratio $\varepsilon_1/\varepsilon_2$. Both are zero outside the gap.

Task 10: Two-dielectric coaxial capacitor

10b · Integrate through both layers

1. The path runs inward, from $\rho = a_2$ to $\rho = a_0$. Along a radial path $\mathrm{d}s = e_\rho\,\mathrm{d}\rho$, and the limits carry the direction, so $E \cdot \mathrm{d}s = E_\rho\,\mathrm{d}\rho$ with ρ decreasing. Insert the two E expressions from 10a:

$$U_{20} = \int_{a_2}^{a_1} \frac{\sigma_0 a_0}{\varepsilon_2 \rho}\,\mathrm{d}\rho + \int_{a_1}^{a_0} \frac{\sigma_0 a_0}{\varepsilon_1 \rho}\,\mathrm{d}\rho$$

2. Each integral of $1/\rho$ gives a logarithm; running inward makes each one negative, for example $\int$ from a_1 to a_0 of $\mathrm{d}\rho/\rho = \ln(a_0/a_1) = -\ln(a_1/a_0)$:

$$U_{20} = -\sigma_0 a_0 \left[\frac{1}{\varepsilon_1}\ln\frac{a_1}{a_0} + \frac{1}{\varepsilon_2}\ln\frac{a_2}{a_1}\right]$$

3. Define the positive geometry/material factor:

$$S = \frac{1}{\varepsilon_1}\ln\frac{a_1}{a_0} + \frac{1}{\varepsilon_2}\ln\frac{a_2}{a_1}$$

4. Rearrange for the inner surface charge density, then substitute it back into D and E:

$$\sigma_0 = -\frac{U_{20}}{a_0 S}; \qquad D(\rho) = -\frac{U_{20}}{S\rho}e_\rho$$

$$E(\rho) = -\frac{U_{20}}{\varepsilon_1 S\rho}e_\rho, \qquad a_0 < \rho < a_1$$

$$E(\rho) = -\frac{U_{20}}{\varepsilon_2 S\rho}e_\rho, \qquad a_1 < \rho < a_2$$

5. **Result:** for positive U_{20} the outer cylinder is at the higher potential, so $\sigma_0 < 0$, $\sigma_2 = -\sigma_0 a_0/a_2 > 0$, and the field points inward, from the outer cylinder to the inner one.

   Check: the field points from high to low potential, and the outer cylinder (node 2) is high when $U_{20} > 0$, so an inward field is right. Integrating the resulting E inward from a_2 to a_0 returns $+U_{20}$. Likely mistake: integrating outward from P_0 to P_2, which flips every sign. The field has units V/m and σ_0 has units C/m².

Task 10: Two-dielectric coaxial capacitor

10c · Capacitance per unit length

1. Capacitance is the charge on the electrode at the higher potential divided by the voltage. For $U_{20} > 0$ that is the outer cylinder, whose facing surface carries $Q = \sigma_2\ 2\pi a_2 \Delta z = -\sigma_0\ 2\pi a_0 \Delta z$:

$$C = \frac{Q}{U_{20}} = \frac{-\sigma_0 2\pi a_0 \Delta z}{U_{20}}$$

2. Substitute $\sigma_0 = -U_{20}/(a_0 S)$. The signs, the voltage and a_0 cancel:

$$C = \frac{2\pi \Delta z}{S}; \quad C_B = \frac{C}{\Delta z} = \frac{2\pi}{S}$$

3. **Result:** C is the capacitance over length Δz; C_B is the capacitance loading per unit length.

   Check: S has units m/F, so C is in farads and C_B in F/m. If $\varepsilon_1 = \varepsilon_2 = \varepsilon$, then $S = \ln(a_2/a_0)/\varepsilon$, recovering the homogeneous coaxial result.

Task 10: Two-dielectric coaxial capacitor

10d · Zero field outside the charged gap

1. For $0 \le \rho < a_0$, a coaxial Gaussian cylinder encloses no free charge: σ_0 lies at $\rho = a_0$, outside this Gaussian surface. Thus its radial flux and field are zero. Within the inner metal wall D is also zero by the conductor condition.

2. For $\rho > a_2$, a Gaussian cylinder outside both facing surfaces encloses their total charge per length:

$$\frac{q_{enc}}{\Delta z} = 2\pi a_0 \sigma_0 + 2\pi a_2 \sigma_2 = 0$$

   The cancellation uses $\sigma_2 = -\sigma_0 a_0/a_2$. Therefore D and E vanish outside the outer conductor; they also vanish within its metal wall.

3. **Result:**

$$E = 0, \quad 0 \le \rho < a_0$$

$$E = 0, \quad \rho > a_2$$

   Check: the applied voltage and field are confined to the annular dielectric gap in this ideal infinitely long, concentric arrangement.

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